

## Web Crawling & Web Scraping - Full Notes

### 1. Web Crawling:

Web crawling means browsing websites automatically to collect data.

### 2. Web Scraping:

Web scraping means extracting specific data from websites.

### 3. Tools:

- requests (HTTP requests)
- BeautifulSoup (HTML parsing)
- lxml (fast parser)
- Selenium (for JavaScript sites)

### 4. Example:

Use requests to fetch page and BeautifulSoup to parse.

### 5. Advanced:

- Headers
- Sessions (login)
- API scraping
- Selenium waits

### 6. Problems:

- NoneType error
- JS not loading
- Blocking

### 7. Best Practices:

- Respect website rules
- Use delay
- Avoid too many requests

### 8. Mini Projects:

- Price tracker

- News scraper

- Job scraper